

AVDASI2 Wind Tunnel Test Plan – Company A Fuselage

(Pod & Payload + HTP & Elevator + Joints & Fairings + VTP & Rudder + Avionics)

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1. Personnel

Suggestions (adapt as your group wishes, you may wish to allocate deputies for workload sharing and resilience):

- **Experimental Aerodynamics Lead (EAL)** – Lachlan Uttley
 - Overall test planning, coordination, and safety
- **Avionics Test Lead (ATL)** - Reuben
 - Avionic systems planning, data logging, avionics contingency planning.
 - Bring Laptop
- **Avionics Technical Operator (ATO)** – Mido Jedda,
 - FCS Installation and control
 - Bring Laptop
- **Wind Tunnel Technical Operator (WTTO)** Ho Lun, (backup Alex Barlow)
 - Tunnel operation on the day (will receive briefing at start of session), tunnel balance data logging.
- **Test Imaging Lead – (TIL)** Ethan Paterson
 - Optical imagery (photos) of assembly in tunnel, key features (smoke, tufts, control surfaces, etc) as planned in advance with stakeholders.
- **Public Engagement Lead (PEL)** – Oliver Schulz
 - Photography of engaging elements of test (coordinate with Test Imaging Lead, but also team photos etc.), draft suggested LinkedIn posts etc.
- **Test Installation Team (TIT) [x4]**– Jad El Badaoui, Mido Jedda, Jack from Pod, Sameer Alhindi

- Installation and removal of assembly in tunnel (note setup/changeover/setdown slots in test schedule). Briefing and support on the day.
- **Test Support Runners (TSP) [x2]**– Alp Tlabar, Alex Barlow
 - Flexible roles on the day, as required. Shouldn't actually be running in QB.
 - Toilet contingency
- **Observers**
 - Wider divisions may wish to observe tests, but we need to avoid overcrowding. ATLS/delegates should monitor space and rota observer attendance if required.

2. Test Pre-requisites & Setup

2.1 Avionics Checklist	Avionics – Failure procedure
Connect CubePilot to laptop via Wi-Fi - should hear buzz noise to confirm connection - if not: proceed through fail checklist Using the UI (User Interface), attempt to move the HTP and VTP - should observe movement in the respective mechanisms - if not: proceed through fail checklist - should observe pitch angle data being logged on laptop - if not: proceed through fail checklist Switch to RC, attempt to move the HTP and VTP - should observe movement in the respective mechanisms - if not: proceed through fail checklist - should observe pitch angle data being logged on laptop - if not: proceed through fail checklist Check For Loose Wiring	- check if Wi-Fi is properly connected on the laptop - check the CubePilot circuit is properly set up ○ does red light on the BEC show? ○ are wires plugged into the correct ports on the cube? - check if Coms in the code are correct ○ "Com7" for USB connection or "udp:0.0.0.0:14550" for WiFi ○ has the correct Com-port been input? - did you connect and disconnect the cube after restarting the UI? - check all JSD connectors

2.2 Fuselage Pre-requisites

Assemble wing fairings onto fuselage

- Ensure nuts tightened sufficiently onto bolt at back of fairing (won't ping off)
- Force balance attachment point accessible for tightening
- Secure nose cone

Assembly HTP & VTP

- HTP and Clamp tightened to fuselage, elevator free to move
- Bottom servo tightened onto clamp
- Elevator mechanism nylock nuts touching linkages but not tight
- VTP secure

Assembly of pod

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Check for loose parts

- Anything loose or malfunctioning may be glued

2.3 Test fittings

Test fittings and mounts found in consumables in M.003

- Ensure all tight and secure on build

Attach Safety wires in tunnel

2.4 Wind Tunnel Risk Assessment to be completed by all present

3. Test schedule

Test Installation – Friday 21 st Feb				
Time	Action	Description	Data Required	Personnel
1440-1520	Assemble Fuselage, bring to tunnel	Arrive at AVDASI lab in time to assemble fuselage before 1500 Put Together HTP, VTP, Fairings and Pod. Sting attachment point tightened on. Check fuselage pre-requisites Check equipment list and bring backup equipment		TIT
1520-1535	Attach force balance and sting	Bring to wind tunnel, same location as A1 lab Attach Main Wing Joint to force balance through fairings. Attach Sting to empennage (will have to be in front of tailplane and rudder) Attach safety wires to build		TIT
1540-1600	Test Operation of mechanisms	Test operation of control surfaces (Elevator and rudder) Check avionics Pre-Requisites		TIT

		Check equipment list and bring backup equipment		
1540-1600	Adjustments	Ensure safety of setup (ask academics) Check calibration of CubePilot Remove sting and adjust internal masses of pod so that Fuselage sits at 0 pitch to save time for Monday. Reattach Sting	CubePilot Angles	TIT
1600	Lab closes			

Static Test – Monday 24th Feb (Sting attached)

Time	Action	Description	Data Required	Personnel
0800-0805	Quick Briefing	Everyone must have Completed wind tunnel risk assessment Have correct safety equipment Recap - Full briefing about Safety, Schedule and Roles must have been read in advance		All
0805-0815	Initial Assembly Proving	Double check installation and operation. Low speed 'Bedding in'		WTTO
0815-0825	Data Verification & calibration	Check CubePilot mounted within pod is measuring correct angles. Ensure Force balance is logging correctly	CubePilot Angles Force logging data	WTTO, ATL
Essential Data Capture				
0825-0835	Flow Speed Test	Find Max Flow speed U_0 Increase flow speed incrementally Is system 'sufficiently robust' (section 4) Fixed Pitch angle @ 0°, Fixed Yaw angle @ 0°, Sting Attached	Max flow speed allowable based on visual stability	All
0835-0845	Test Yaw Angles	Test different Yaw angles Test Yaw @ 0°, 1°, 2° Is system 'sufficiently robust' (section 4) Fixed Flow Speed U_0 Fixed Pitch angle @ 0°, Sting Attached	Wind tunnel Force Logging Data	All
0850-0905	Test Pitch Angles	Test different Pitching angles Test Pitch angles in 5° increments @ -10°, -5°, 5°, 10°, 15°, 20° Is system 'sufficiently robust' (section 4) Fixed Flow Speed U_0	Wind tunnel Force Logging Data	All

		Fixed Yaw angle @0°, Sting Attached		
0905-0910	Rudder deflection speed	Does Rudder meet required deflection speed and angles under load TIL video and imaging Fixed Flow Speed U_0 Fixed Yaw angle @0°, Fixed Pitch angle @ Sting Attached	Realtime Video	All TIL
0905-0910	Test Rudder deflections	Rudder at deflection 10°, 25°, 35, 40° Observe moments <i>If behind Schedule only test at full deflection</i> Fixed Flow Speed U_0 'ONLY at' Fixed Yaw angle @0°, Fixed Pitch angle @0°, Sting Attached	Wind tunnel Force logging data time@ 10:02:25 Negative also	All
0910-0920	Elevator speed and deflections	Elevator at deflections 10°, 20°, 30°, 40, 45 Observe moments Fixed Flow Speed U_0 'ONLY at' Fixed Yaw angle @0°, Fixed Pitch angle @0°, Sting Attached	Time 10:05:40 Negative also	
0920-0930	No Fairings test	Drag and smoke visualisation? Fairings Removed , Record changes Fixed Flow Speed U_0 Fixed Yaw angle @0°, Fixed Pitch angle @0°, Sting Attached	Pitch angles from Cube Wind tunnel Force logging data (Video)	All
0920-0930	Static test – Dynamic test change over	Remove sting from empennage Readjust internal masses in pod so that fuselage sits at 0° pitch Keep Fairings off		TIT

Dynamic Test – Monday 24 th Feb				
Time	Action	Description	Data Required (Expected)	Personnel
			already stable in pitch Manual pitch disturbance damped return	

0945-0955	Elevator deflections	Elevator locked at deflections by servo, see what happens to pitch Test at low deflections to avoid slamming into floor Fixed Flow Speed U_0 Fixed Yaw angle @0°, Sting Not attached	Pitch angles from Cube Wind tunnel Force logging data Order of deflections -5° -10 0 -10 redone 0 -20 0 -30 0 10 0 20 0 30	All
0955-1005	Fuselage deflection	PID tuning Hold fuselage at large pitch angle (big stick) then release. Visually see response Fixed Flow Speed U_0 Fixed Yaw angle @0°, Sting Not attached Control Surfaces @0° deflection	Pitch angles from Cube Wind tunnel Force logging data	All
1010-1040	PID Avionics System test	Activate automatic control system Elevators should adjust to conditions Should provide greater stability in wind Attempt to deflect? (big stick) Test Minimum Pitch Range (The UAV shall achieve a minimum pitch trim range ($a_{max}-a_{min}$) of 25° at a test speed of 10m/s) Fixed Flow Speed U_0 Fixed Yaw angle @0°, Sting Not attached	Pitch angles from Cube Wind tunnel Force logging data Realtime video	ATL ATO All
1040-1100	Smoke Visualisation test	Add smoke, observe flow over surfaces. TIL imaging Fixed Flow Speed U_0 Fixed Yaw angle @0°, Sting Not attached	Pitch angles from Cube Wind tunnel Force logging data Realtime video	All

		Automatic flight @0 pitch		
1100	Changeover with company B	Remove Build from tunnel Ensure all parts are collected and tunnel left tidy		TIT

3.2 If Time to spare

Yaw and pitch angle static test to higher resolution

- yaw every half°
- pitch every 3°

4. Data

4.1 Data Acquisition

- Do not delete any data even if it is believed to be 'bad data'
 - o Ensure everything is logged
- WTTTO will be briefed on the day on how to logging of force drag and moment
- ATO and ATL responsible for logging pitch and control surface deflections
 - o Connect USB to mission planner, download data flash logs after each test where it is required

4.2 Relevant Spec

4.2.1 Fuselage

- The UAV shall achieve a minimum pitch trim range ($\alpha_{max}-\alpha_{min}$) of 25° at a test speed of 10m/s
- The UAV shall achieve for rudder deflection $dC_{yaw} force d\delta r > 2.3$ and for yaw angle $dC_{yaw} force d\zeta > 2$
- The design of the pod shall be optimised to reduce aerodynamic drag. The drag shall be predicted via calculation. Pod load data shall be used to confirm the Pods aerodynamic performance.

4.2.2 Fairings

- Fairings shall be removeable, where practical, to enable their aerodynamic contribution to be isolated during Wind Tunnel testing.
- The fairings, as described in 2.6, shall reduce aerodynamic drag of the UAV at all flight speeds and angles of attack.

4.2.3 Static

- Systems shall be sufficiently robust to withstand, without requiring emergency/reactive modifications, a full range of rotation, while the fuselage is held fixed at a pitch angle of zero degrees, under wind tunnel testing conditions of wind speeds up to 20 meters per second.

4.2.4 Sensing

- Flight data, including but not limited to wind speed (from wind tunnel equipment), aerodynamic loads (from tunnel balance, fuselage pitch angle (from FCS and/or tunnel), elevator and rudder angles (from mechanism conversion

curve or selected device), pitch control behaviour, and static pressure (from wind tunnel and wing taps (differential)) shall be logged

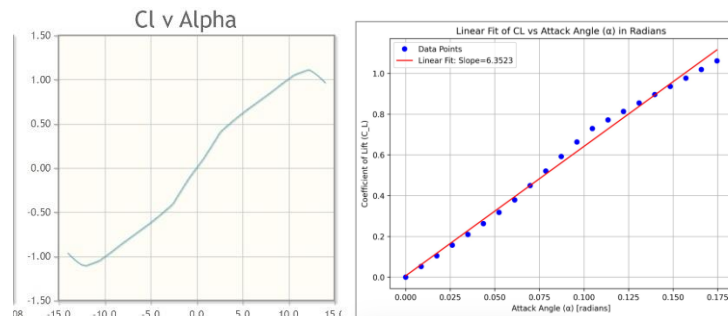
- Logging of all data should be at minimum 5Hz and ideally at >20Hz

4.2.4 Control

- The elevator shall be capable of deflection within an angle range of [-45,+45]° degrees with respect to the HTP aerofoil mean line.
- The elevator shall be capable of achieving the commanded positions within one degree under loading conditions up to the ultimate design load
- The elevator shall transition between maximum and minimum elevator angle less than 1.5 seconds.
- Pitch angle demand via TMS in 'automatic' flight using a PID controller. The controller shall be able to automatically adjust the elevator deflection to achieve the commanded pitch angle.
- The rudder shall be capable of rotation within an angle range of [-40,+40]° degrees with respect to the VTP aerofoil mean line.
- The rudder shall be capable of achieving the positions to within one degree of demanded angle, under loading conditions up to the ultimate design load.
- The rudder control system shall provide rudder angle demand via RCS in 'manual flight'.

4.3 Expected data

4.3.1 yaw angle static test. VTP is using NACA 0012. Polars using approximate Re value of 2×10^5 are below on left and a graph plotted by VTP of small angles of attack on the right



Expecting roughly linear increase in Cl at low Angles of Attack correlating to a linear increase in yawing moments as yaw increases.

Predicted rate of change of yawing moment with respect to yaw angle:

$$\frac{\partial C_{yaw force}}{\partial \zeta} = 3.2$$

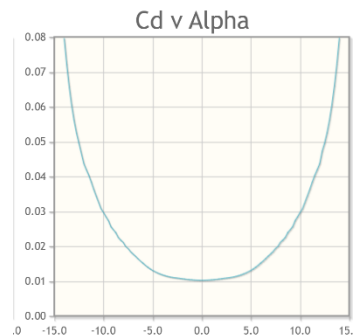
(Per radian)

VTP ac at 875mm from centre giving

$$\frac{\partial C_{yaw moment}}{\partial \zeta} = 2.8$$

(Per radian)

As yaw increases expecting slight increase in drag at low angles.



4.3.2 Pitch angle static test

HTP is also using NACA 0012 airfoil. Polars using approximate Re value of 2×10^5 :



As Pitch increases expecting roughly linear increase in lift produced by tailplane until separation at around 10° where it is expected to curve off. This correlates to a linear and small increase in lift and a larger increase in pitching moment before separation. HTP team predicted maximum Cl and therefore pitching moment at around 10° AoA

Tailplane lift gradient calculated as:

$$a_1 = \frac{dC_l}{d\alpha} = 3.88$$

(per radian)

Tail Plane around 860mm from force balance

$$\frac{dC_{pitching\ moment}}{d\alpha} = 0.835$$

(per radian)

4.3.3 Rudder deflection

Calculated rate of change of with respect to rudder deflection by VTP team:

$$\frac{\partial C_{yaw\ force}}{\partial \delta} = 2.304$$

$$\frac{\partial C_{yaw\ moment}}{\partial \delta} = 2.016$$

Predicted linear relationship at small angles, expect it to curve off at larger angles of deflection when there will be flow separation

4.3.4 Elevator Deflection Dynamic test

Calculated rate of change of lift of tailplane and therefore pitching force. Expect linear relationship between force and therefore pitching moment in at small deflections before separation

Pitch angles assumed to be linear to pitching force.

5. Equipment

Fuselage build equipment	Avionics Equipment
- Fuselage - Main wing joint - Wing fairing - o Bolts and nuts.. - Nose Cone - Empennage joint - o 2xbolts through fuselage with wing nuts - o 3xfairings to clamp - HTP - o Tailplane - o Mechanism attached to bottom joint under fuselage - VTP - Test fittings found in consumables (M.003) - o 6dof force balance mount - o Sting attachment point (no spares do not lose) - Backup Equipment and tools - o Main wing joint fastener - o Spare bolts and nuts for build Tape & glue	- Laptop x2 - o charger - CubePilot + circuit - o BEC circuit - o Cube power cable - o Wi-Fi chip - o RC chip - RC controller - Long emergency USB cable - Power cable - Backup equipment - o Wires - o wire strippers - o Tape

5.3 Equipment provided by staff

- Wind tunnel, 7x5ft working section, including 6dof balance, measuring system, and appropriate attachments.
- Power supply (single variable voltage), e-stop, wires sufficient to run from test control table into tunnel. XT60 female connector at tunnel end.
- USB extension (USB-A female to male), 5m powered – sufficient from tunnel to table.
- Briefings and supervision as appropriate, but student teams will lead own tests.

6. Safety

Everyone present to read and complete risk assessment, published in advance

Be aware of your surroundings, you are responsible for your own build

The Wind Tunnel Lab is a professional research environment, all student-researchers are responsible for their safety and conduct, and the safety of others.

Risk assessment will be published in advance, all members to read, query where necessary, raise any additions as required.

Tests will be conducted under divisional leadership arrangements, however any necessary staff instruction will supersede these and must be promptly heeded. Staff will permit team autonomy where appropriate, but may intervene to support safety or effective outcomes.

7. Photography & other elements

- Photography of smoke visualisation of flow
- Photography of tufts
 - Points of separation
 - Changes in behaviour after changes in setup
- Essential Test photography and videography that can't be logged
 - required deflections and speeds
 - Stability at flow speed
- LinkedIn photography